

BS – 101 Engineering Physics

Lecture – 10

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Magnetic Field and Magnetic force

- Magnetic phenomena were first observed at least 2500 years ago in fragments of magnetized iron ore found near the ancient city of Magnesia (now Manisa, in western Turkey). These fragments were what are now called **permanent magnets**. Permanent magnets were found to exert forces on each other as well as on pieces of iron that were not magnetized



Magnetism (Contd.)

- It was discovered that when an iron rod is brought in contact with a natural magnet, the rod also becomes magnetized.
- When such a rod is floated on water or suspended by a string from its center, it tends to line itself up in a north-south direction.
- The needle of an ordinary compass is just such a piece of magnetized iron.



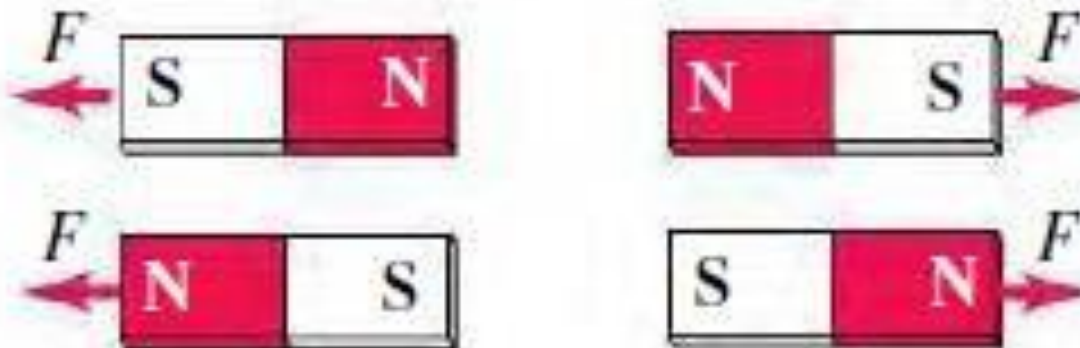
Magnetic Poles

- Before the relationship of magnetic interactions to moving charges was understood, the interactions of permanent magnets and compass needles were described in terms of *magnetic poles*.
- If a bar-shaped permanent magnet, or *bar magnet*, is free to rotate, one end points north. This end is called a *north pole* or *N pole*; the other end is a *south pole* or *S pole*.
- Opposite poles attract each other, and like poles repel each other

Magnetic Poles (Figure)



(b) Like poles repel.

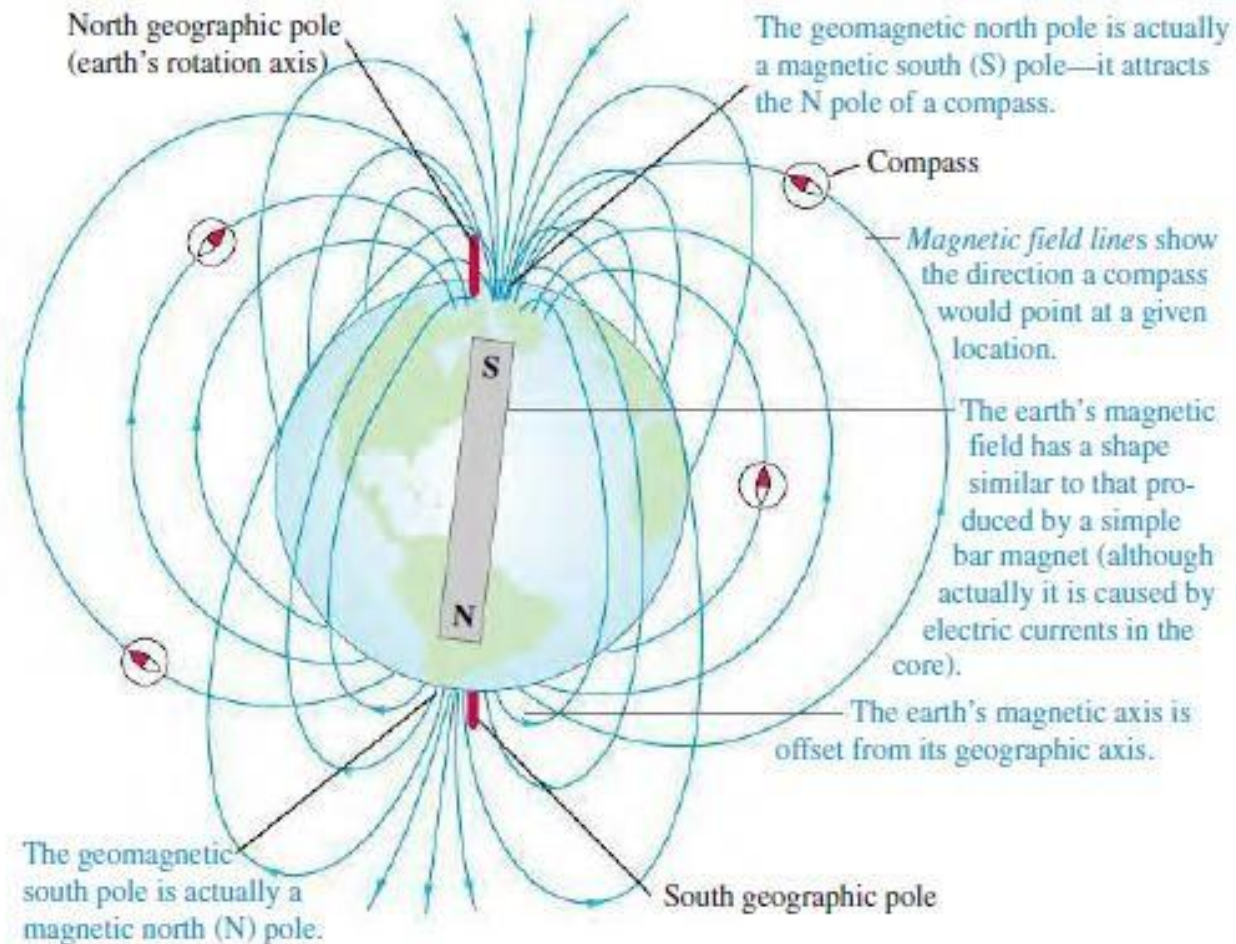




Magnetic Poles (Earth)

- The earth itself is a magnet. Its north geographic pole is close to a magnetic *south* pole, which is why the north pole of a compass needle points north.
- The earth's magnetic axis is not quite parallel to its geographic axis (the axis of rotation), so a compass reading deviates somewhat from geographic north.
- This deviation, which varies with location, is called *magnetic declination* or *magnetic variation*.
- Also, the magnetic field is not horizontal at most points on the earth's surface; its angle up or down is called *magnetic inclination*.
- At the magnetic poles the magnetic field is vertical.

Magnetic Poles (Earth Figure)





Magnetic Poles (Double)

- The concept of magnetic poles may appear similar to that of electric charge, and north and south poles may seem analogous to positive and negative charges. But the analogy can be misleading.
- While isolated positive and negative charges exist, there is *no* experimental evidence that one isolated magnetic pole exists; poles always appear in pairs.



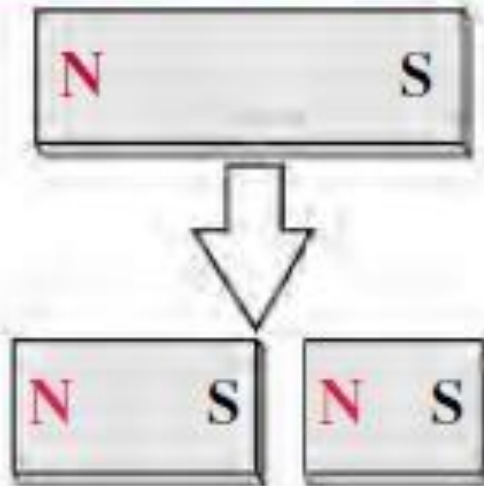
Magnetic Poles (Double) Contd.

- If a bar magnet is broken in two, each broken end becomes a pole.
- The existence of an isolated magnetic pole, or **magnetic monopole**, would have sweeping implications for theoretical physics.
- Extensive searches for magnetic monopoles have been carried out, but so far without success.

Magnetic Poles (Double) Contd.

In contrast to electric charges, magnetic poles always come in pairs and can't be isolated.

Breaking a magnet in two ...



... yields two magnets,
not two isolated poles.



Magnetic Field

- To introduce the concept of magnetic field properly, let's review our formulation of *electric* interactions in previous lectures, where we introduced the concept of *electric* field. We represented electric interactions in two steps:
 - 1. A distribution of electric charge creates an electric field $\mathbf{E}$ in the surrounding space.
 - 2. The electric field exerts a force $\mathbf{F} = q\mathbf{E}$ on any other charge q that is present in the field.



Magnetic Field (Contd.)

- We can describe magnetic interactions in a similar way:
- 1. A moving charge or a current creates a **magnetic field** in the surrounding space (in addition to its *electric* field).
- 2. The magnetic field exerts a force **F** on any other moving charge or current that is present in the field.



Magnetic Field (Direction)

- Like electric field, magnetic field is a *vector field*—that is, a vector quantity associated with each point in space. We will use the symbol **B** for magnetic field.
- At any position the direction of **B** is defined as the direction in which the north pole of a compass needle tends to point.



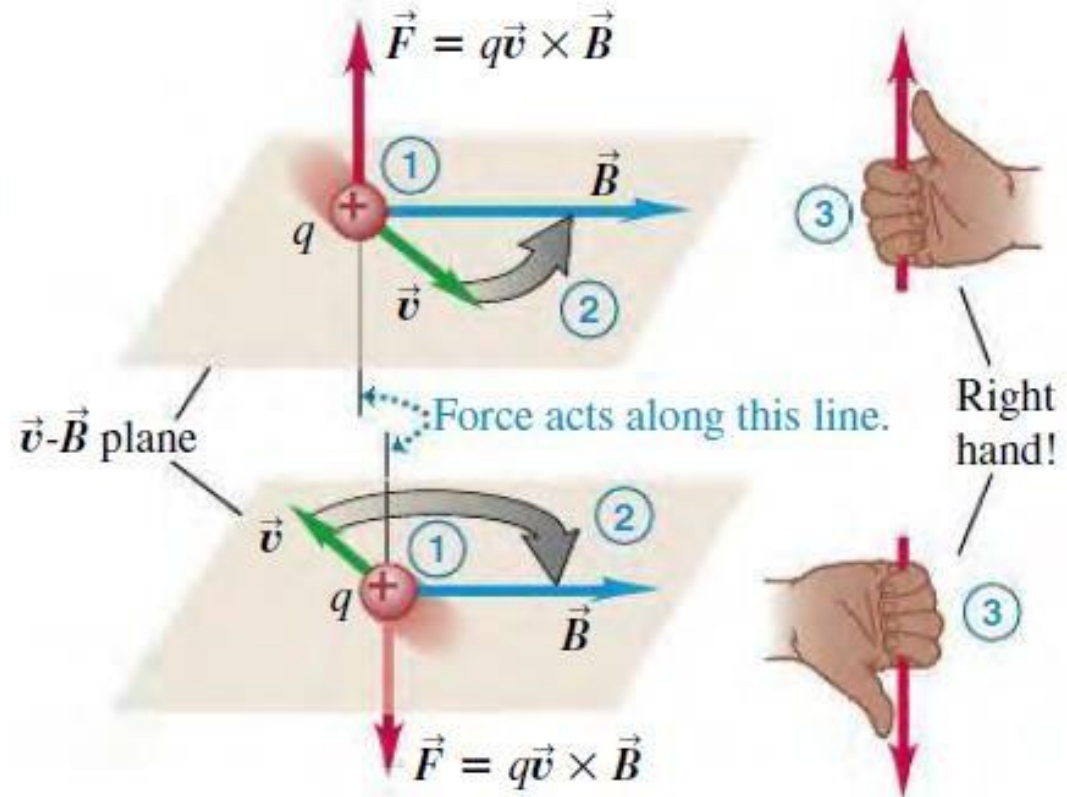
Magnetic Forces on Moving Charges

- $F = |q| B v \sin(\varphi)$
- Force is proportional to the magnitude of charge.
- Force is proportional to the magnitude of magnetic field.
- Force is proportional to the magnitude of velocity.
- $\vec{F} = q (\vec{v} \times \vec{B})$

Direction of Magnetic Force

Right-hand rule for the direction of magnetic force on a positive charge moving in a magnetic field:

- ① Place the $\vec{v}$ and $\vec{B}$ vectors tail to tail.
- ② Imagine turning $\vec{v}$ toward $\vec{B}$ in the $\vec{v}$ - $\vec{B}$ plane (through the smaller angle).
- ③ The force acts along a line perpendicular to the $\vec{v}$ - $\vec{B}$ plane. Curl the fingers of your *right hand* around this line in the same direction you rotated $\vec{v}$. Your thumb now points in the direction the force acts.



Magnetic Field (Unit)

- The *units* of B must be the same as the units of $\frac{F}{qv}$.
- Therefore the SI unit of B is equivalent to $1 \frac{Ns}{Cm}$ or, since one ampere is one coulomb per second ($1 A = 1 \frac{C}{s}$), $1 \frac{N}{Am}$. This unit is called the **tesla** (abbreviated T), in honor of Nikola Tesla (1856–1943), the prominent Serbian- American scientist and inventor:

$$1 \text{ T} = 1 T = 1 \frac{N}{Am}$$

- Another unit of B , the **gauss** ($1 G = 10^{-4}T$), is also in common use. The magnetic field of the earth is of the order of $10^{-4}T$ or $1 G$.

Magnetic Flux

Magnetic flux through a surface $\Phi_B = \int B \cos \phi \, dA = \int B_{\perp} \, dA = \int \vec{B} \cdot d\vec{A}$

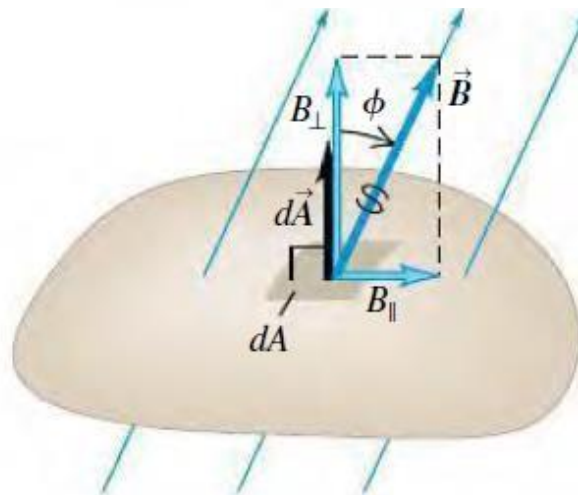
Magnitude of magnetic field $\vec{B}$

Component of $\vec{B}$ perpendicular to surface

Angle between $\vec{B}$ and normal to surface

Element of surface area

Vector element of surface area





Magnetic Flux (Scalar)

- Magnetic flux is a *scalar* quantity. If $\mathbf{B}$ is uniform over a plane surface with total area A , then B and ϕ are the same at all points on the surface, and

$$\phi_B = BA \cos(\phi)$$

- If $\mathbf{B}$ is also perpendicular to the surface (parallel to the area vector), then $\cos \phi = 1$ and above eq. reduces to $\phi_B = BA$.

Magnetic Flux (Unit)

- The SI unit of magnetic flux is equal to the unit of magnetic field ($1\ T$) times the unit of area ($1\ m^2$). This unit is called the **weber** ($1\ Wb$), in honor of the German physicist Wilhelm Weber (1804–1891):

$$1\ Wb = 1\ Tm^2$$

- Also, $1\ T = 1\ \frac{N}{Am}$, so

$$1\ Wb = 1\ Tm^2 = 1\ \frac{N}{Am}$$

Gauss's Law for Magnetism

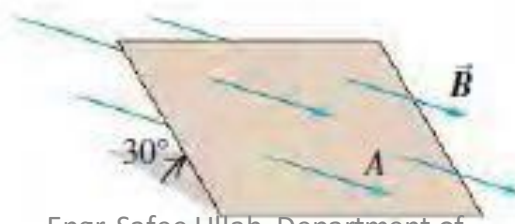
- In Gauss's law the total *electric* flux through a closed surface is proportional to the total electric charge enclosed by the surface. For example, if the closed surface encloses an electric dipole, the total electric flux is zero because the total charge is zero. By analogy, if there were such a thing as a single magnetic charge (magnetic monopole), the total *magnetic* flux through a closed surface would be proportional to the total magnetic charge enclosed. But we have mentioned that no magnetic monopole has ever been observed, despite intensive searches. This leads us to **Gauss's law for magnetism:**

$$\oint \vec{B} \cdot d\vec{A} = 0$$

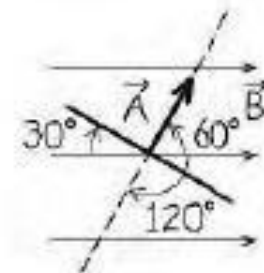
Magnetic Flux Calculations

- **Given Figure** is a perspective view of a flat surface with area 3.0 cm^2 in a uniform magnetic field $\mathbf{B}$. The magnetic flux through this surface is $+0.90 \text{ mWb}$. Find the magnitude of the magnetic field and the direction of the area vector $\mathbf{A}$. The angle between area $\mathbf{A}$ and magnetic field $\mathbf{B}$ is 60° .

(a) Perspective view

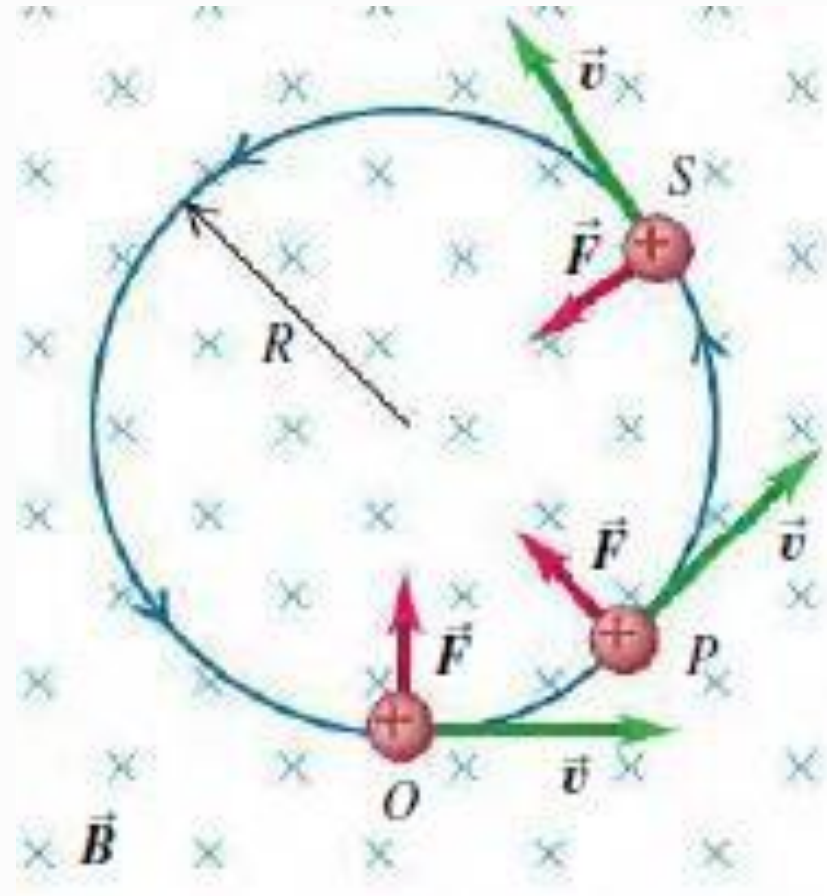


(b) Our sketch of the problem
(edge-on view)



Motion of Charged Particles in a Magnetic Field

- $F = |q|B = m \frac{v^2}{R}$
- $R = \frac{mv}{|q|B}$
- $\omega = \frac{v}{R} = v \frac{|q|B}{mv} = \frac{|q|B}{m}$





Electron Motion in a Magnetron

- A magnetron in a microwave oven emits electromagnetic waves with frequency $f = 2450$ MHz. What magnetic field strength is required for electrons to move in circular paths with this frequency?

- $m_e = 9.11 \times 10^{-31} \text{ kg}$

- $q_e = 1.6 \times 10^{-19} \text{ C}$

Helical Particle Motion in a Magnetic Field

- In a situation like that shown in previous figure, the charged particle is a proton ($q_p = 1.6 \times 10^{-19} C$, $m_p = 1.67 \times 10^{-27} kg$) and the uniform, $0.500 - T$ magnetic field is directed along the x -axis. At $t = 0$ the proton has velocity components $v_x = 1.50 \times 10^5 m/s$, $v_y = 0$, and $v_z = 2.00 \times 10^5 m/s$. Only the magnetic force acts on the proton. (a) At $t = 0$, find the force on the proton and its acceleration. (b) Find the radius of the resulting helical path, the angular speed of the proton, and the *pitch* of the helix (the distance traveled along the helix axis per revolution).